2026 Artificial Intelligence and Space Computing Challenge

Space Intelligence Empowering Zero Hunger Track - Task Description of Final Round

1. Task

Participants need to utilize multi-temporal Sentinel-2A/2B multi-spectral surface reflectance imagery and auxiliary phenology data (MCD12Q2) within the rice growing season to simultaneously predict for hidden image points in the test set: 1) Crop category (distinguishing between non-agricultural background, rice, corn, and soybean); 2) Specific phenological stage nodes (a total of 7 stages from the start of the growing season to maturity).

2. Data Format

2.1 Training Set

The training set mainly consists of two parts: remote sensing image data and label data.

- region_train_1/, region_train_2/, region_train_3/, and region_train_4/: Multi-temporal Sentinel-2 remote sensing imagery data for training regions. File naming format: regionXX_YYYY-MM-DD-00-00_YYYY-MM-DD-23-59_Sentinel-2_L2A_Bxx_(Raw).tiff

  Where:

  - regionXX represents the region number, such as region00, region52
  - YYYY-MM-DD represents the image date
  - Bxx represents the band number

The training set usually contains multiple band files for the same region and date, mainly including: B01, B02, B03, B04, B05, B06, B07, B08, B8A, B09, B11, B12.

Each band is stored as an individual .tiff file, which together constitute a multi-spectral image for the corresponding date.

- points_train_label.csv: Training sample point label file

  The training label file points_train_label.csv is stored in CSV format,

where each line represents a labeled training sample with fields: point_id, Longitude, Latitude, phenophase_date, crop_type, phenophase_name

The meaning of each field is as follows:

- point_id: Sample point number
- Longitude: Sample point longitude
- Latitude: Sample point latitude
- phenophase_date: Observation date for the sample point
- crop_type: Crop type label, such as rice, corn, soybean
- phenophase_name: Phenology label, such as Greenup, MidGreenup, Peak, Maturity, Senescence, MidSenescence, Dormancy

The training samples are used as follows: based on the coordinates, crop type, and phenology date in points_train_label.csv, find the multi-band imagery of the corresponding region and date in regionxx_train, and extract spectral information from the sample point location to model crop type identification and phenological stage classification.

2.2 Test Set

With the start of the Final Round, the test dataset will no longer be publicly released. The format of the test set is the same as that of the training set.

Participants are required to submit a Docker image containing their model code and upload it to the zero2x competition server for evaluation.

The submitted results must include **predictions for all test points; otherwise, no score will be awarded.**

Test set input path: e.g., /input/ (mounted by the platform)

```
Track1
/input/
|-- points_test.csv
`-- region_test\
      `--region03_2018-07-13-00-00_2018-07-13-23-59_Sentinel-2_L2A_B01_(Raw)
      ……
      `--region03_2018-07-13-00-00_2018-07-13-23-59_Sentinel-2_L2A_B12_(Raw)
   `-- (Bulk Sentinel-2 .tiff files, named by region+date+band)
```

2.3 Model Inference Result Format

The prediction result should be a single JSON file placed in the /output directory and named **result.json.**

The file should follow the format below:

```
{
  "{Longtitude}_{Latitude}_{Date}": [
    "{CropType}",
    "{PhenophaseStage}"
  ],
  ...
}
```

Output format example:

```
{
  "124.703696_48.543523_2018/9/1": [
    "corn",
    "Senescence"
  ],
  ……

}
```

3. Scoring Criteria

The algorithmic objective score is determined by Crop Classification Macro-average F1 (40% weight) and Rice Phenology-specific Macro-average F1 (60% weight). The background scoring system will perform line-by-line comparison between the CSV file submitted by participants and the hidden ground truth.

$$Algorithm_Score = (0.4 \times Macro - F1_{Crop} + 0.6 \times Macro - F1_{RicePheno}) \times 100$$

3.1 Crop Classification Score ($Macro - F1_{Crop}$ (Macro−F1_Crop , disclosed indicator, weight 40%)

Evaluates the model's overall identification and background stripping ability across 3 major categories (rice, corn, soybean). Calculate the F1 score for each category independently and then take the arithmetic mean:

$$Macro - F1_{Crop} = \frac{1}{3}\sum_{c=1}^{3} \mathrm{F1}_{\mathrm{Crop_c}}$$

Formula parameters and calculation notes:

c : Represents specific crop category index, $c \in \{1,2,3\}$ (strictly corresponding to: 1-rice, 2-corn, 3-soybean).

$F1_{Crop_c}$: The model's exclusive $F1$ score on target class c, calculated by

the harmonic mean of precision （$Precision_c$） and recall （$Recall_c$） of that class, formula is:

$$F1_{Crop_c} = \frac{2 \times Precision_c \times Recall_c}{Precision_c + Recall_c}$$

Where, the precision and recall for class cc are defined as follows:

Precision ($Precision_c$): Among all samples predicted by the model as class cc, the proportion whose actual label is indeed class cc.

$$Precision_c = \frac{TP_c}{TP_c + FP_c}$$

Recall ($Recall_c$): Among all actual class cc samples in the test set, the proportion correctly predicted by the model.

$$Recall_c = \frac{TP_c}{TP_c + FN_c}$$

The confusion matrix elements used to calculate the above indicators are defined as follows (Note: this assessment only verifies crop classification accuracy and does not consider phenology prediction results):

TP_c (True Positives): The number of samples where the actual crop category is c, and the model's predicted crop category is also cc.

FP_c （(False Positives / Misreport): The number of samples where the actual crop category is not c, but the model incorrectly predicts its crop category as c.

FN_c(False Negatives / Omission): The number of samples where the actual crop category is c, but the model fails to predict it as c (misjudged as other crops or background).

(Note: During automated program evaluation, if a category's TP+FP=0, the Precision for that category is recorded as 0; if TP+FN=0, the Recall for that category is recorded as 0; if Precision + Recall = 0, the F1_Crop for that category is recorded as 0.)

3.2 Rice Phenology-specific Classification Score ($Macro\text{−}F1_{RicePheno}$ Macro−F1_RicePheno , disclosed indicator, weight 60%)

The core decisive indicator for this challenge. To ensure the business goal of rice extraction is not diluted, this indicator strictly evaluates the phenological stage extraction accuracy of the rice category only (phenology prediction errors for corn

and soybean are not included in this item). Prediction results must achieve "Double Hit":

$$Macro - F1_{RicePheno} = \frac{1}{7}\sum_{i=1}^{7} \mathrm{F1}_{\mathrm{Rice_i}}$$

Formula parameters and "Double Hit" calculation description:

i : Represents specific rice phenological stage index, $i \in \{1,2,3,4,5,6,7\}$ (strictly corresponding from T_1 Growing Season Start to T_7 Maturity).

$F1_{Rice_i}$: The model's exclusive $F1$ score on the specific joint target of accurately extracting "rice at phenological stage i". Calculated by the harmonic mean of the target's precision ($Precision_{Rice_i}$) and recall ($Recall_{Rice_i}$):

$$F1_{Rice_i} = \frac{2 \times Precision_{Rice_i} \times Recall_{Rice_i}}{Precision_{Rice_i} + Recall_{Rice_i}}$$

Where, the formulas for precision and recall are:

Precision ($Precision_{Rice_i}$): Among all samples predicted by the model as "rice and phenology stage i", the proportion where the actual situation is indeed so.

$$Precision_{Rice_i} = \frac{TP_i}{TP_i + FP_i}$$

Recall ($Recall_{Rice_i}$): Among all actual "rice and phenology stage i" samples in the test set, the proportion completely correctly identified by the model.

$$Recall_{Rice_i} = \frac{TP_i}{TP_i + FN_i}$$

To prevent the model from cheating by using background loopholes or single-task over-fitting, the confusion matrix elements (TPi, FPi, FNi) in the above formula must satisfy extremely strict "Exact Match" conditions, i.e., crop category and phenological stage must be predicted correctly simultaneously:

- TP_i(True Positives / Perfect Double Hit):

  The actual crop category is rice AND the actual phenological stage is i; simultaneously, the model's predicted crop category is rice AND the predicted phenological stage is also i. (The double labels must be completely consistent with the ground truth).

- FP_i(False Positives / Misreport Penalty):

  The model predicts the point is "rice at phenological stage i", but it is not actually quite so. A misreport is recorded if any of the following occur:

  - Wrong crop (Spatial Misextraction): The actual ground cover is not rice at all

(but corn, soybean, or background), but the model incorrectly predicts it as "rice" and forcibly assigns phenological stage i.

- Wrong stage (Temporal Misjudgment): The actual ground cover is indeed rice, but it is actually at phenological stage j ($j \neq i$), while the model incorrectly predicts it at phenological stage i.

- FN_i(False Negatives / Omission Penalty):

The actual nature of the point is "rice at phenological stage i", but the model fails to provide a perfect double prediction. An omission is recorded if any of the following occur:

- Missed crop (Spatial Omission): The point is indeed rice, but the model does not recognize it as rice, misjudging it as other crops or background (in this case, regardless of the predicted phenological stage, it counts as an omission).
- Wrong stage (Temporal Omission): The point is rice, and the model correctly recognizes it as rice, but misjudges its phenological stage as j（$j \neq i$） or predicts it as having no phenology (*0*).

(Note: During automated program evaluation, if TPi+FPi=0, then the item's Precision is recorded as 0; if TPi+FNi=0, then the item's Recall is recorded as 0; if Precision + Recall = 0, then the F1 for that specific phenological stage is recorded as 0.)

3.4 Application Implementation Scheme Scoring

Dimension	**Score**	**Scoring Key Points**
Part I: Overall Task Implementation Architecture	10	* Is the full-process logic clear and complete, covering all stages from **data injection** to **result downlink**? * Does the technical route incorporate key designs such as **modularity, heterogeneous scheduling, and fault tolerance**? * Is the architecture design reasonable and adaptable to **resource-constrained space environments**?
Part II: On-Orbit Feasibility	15	* Are the **uplink file sizes** and **compression strategies** clearly defined? * Are there reasonable estimates for resource

Dimension	Score	Scoring Key Points
		consumption, including **CPU/GPU, memory, and storage**? * Are the types of dependent data sources and their management methods specified? * Is there an analysis of single-execution time and **worst-case latency**?
Part III: Innovation of Implementation Path	20	* Does it involve key technical breakthroughs (e.g., **model compression, lightweight inference, distributed computing**)? * Does it clearly explain the fundamental differences from ground-based solutions? * Does it feature targeted designs for **space-specific constraints** (power consumption, radiation, autonomous operation)?
Part IV: Value and Application Scenarios	20	* Is the path for value transformation (commercial or social benefits) clearly defined? * Are the application scenarios specific and relevant to real-world production and daily life? * Does it demonstrate an extended application model for **Satellite-Terrestrial Coordination**?
Part V: Future Roadmap	15	* Does it clearly articulate the application prospects? * Are there proposed paths for technical improvement? * Does it identify key development milestones for **on-orbit computing** in this specific field?
Part VI: Team Introduction	5	* Does the team background match the project requirements? * Does it demonstrate the team's relevant experience and capabilities in this field?
Presentation and Logic	15	* Is the PPT structure clear, with concise and powerful delivery? * Is the technical language accurate and the logic

Dimension	Score	Scoring Key Points
		coherent? * Does it highlight project highlights and **differentiated advantages**?